

WEEK 1



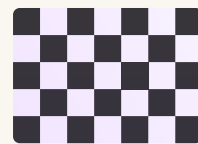
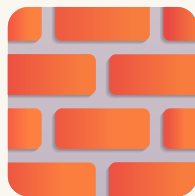
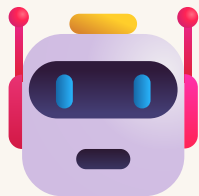
While Loops

Move until you hit a wall

벽에 닿을 때까지 계속 이동하기 (영어코딩)

BIG PICTURE

8 weeks → solve any maze 🌀



This week: keep moving forward until a wall stops you.

반복문으로 미로를 스스로 풀어요

HOW TO COUNT

Count the axis — one block, one number



- 1 Stand on the start block — that spot is $(0, 0, 0)$.
- 2 $x \rightarrow$ walk **right**. Add **+1** for every block.
- 3 $y \rightarrow$ jump **up**. Add **+1** for every block.
- 4 $z \rightarrow$ walk **forward**. Add **+1** for every block.

Say them in order every time: $(x, y, z) = \text{across, up, forward}$.

한 칸 움직이면 +1 — 오른쪽 x , 위로 y , 앞으로 z . 순서는 늘 (x, y, z)

TYPE IT IN

Put the numbers into Minecraft



^x
across →

^y
up ↑

^z
forward ↗

```
place STONE at (x, y, z)
```

- ✓ Open **Code Builder** (press **C** in the world)
- ✓ Type the numbers in order — **(x, y, z)**
- ✓ Press **Run ▶** — your block appears at that spot

코드 빌더(C) 열고 → (x, y, z) 순서로 숫자 넣고 → Run ▶

CONCEPT

What is a **while** loop?

It repeats the body **again and again** while a condition is **true**.

```
while no wall ahead:  
    move forward
```

Check → run → check again. Stop the moment it's false.

조건이 참인 동안 계속 반복해요

CONCEPT

while vs writing it many times



while loop

stops exactly
at the wall



copy-paste moves

guesses the count
(too far / too short)

반복문은 벽 앞에서 정확히 멈춰요

WATCH IT!

The loop plays out

```
steps = 0
```

```
no wall ahead?
```

```
agent.move(FORWARD, 1)
```

```
▸ Agent
```

CODE

Detect the wall

```
# True when a block is right in front  
agent.detect(AgentDetection.BLOCK, FORWARD)
```

`agent.detect(...)` answers a yes/no question: is there a block ahead?

앞에 벽이 있는지 알려줘요

CODE

Move until the wall

```
while not agent.detect(AgentDetection.BLOCK, FORWARD):  
    agent.move(FORWARD, 1)
```

not flips it: loop **while there is NO wall** ahead.

벽이 없는 동안 한 칸씩 앞으로

CODE

Two loops + a turn

```
while not agent.detect(AgentDetection.BLOCK, FORWARD):  
    agent.move(FORWARD, 1)  
agent.turn(RIGHT_TURN)  
while not agent.detect(AgentDetection.BLOCK, FORWARD):  
    agent.move(FORWARD, 1)
```

Reach the wall, turn, then keep going down the new path.

PREDICT 🤖

When does it stop?

Predict first — then click reveal

```
while not agent.detect(BLOCK, FORWARD):  
    agent.move(FORWARD, 1)
```

Reveal output 🔍 reveal

VOCAB

New words today

while loop repeat a body while a condition stays true

condition a yes/no test the loop checks each time

detect sense whether a block is there

not flips true to false and false to true

반복문

조건

감지하다

부정 (반대로)



DEBUG

What's wrong with this code?

Goal: move forward **until** a wall.

buggy

```
while agent.detect(BLOCK, FORWARD):  
    agent.move(FORWARD, 1)
```

Think first — then click next.



The bug

Bug: the condition is backwards. This loops *while there IS a wall* — so it never even starts. Add **not** to loop while the path is clear.

fixed

```
while not agent.detect(BLOCK, FORWARD):  
    agent.move(FORWARD, 1)
```

YOUR TURN!

Reach the wall

- ✓ Open the Week 1 world + Code Builder
- ✓ Use a `while not ... detect` loop
- ✓ Move the agent to the end of the straight maze
- ✓ It should stop right in front of the wall
- ✓ Send a video of the run on KakaoTalk



QUIZ

QUESTION 1 OF 3

What does a `while` loop do?

Only turns the agent.

Places a block each time.

Runs its body exactly once.

Repeats its body while the condition is true, then stops.



QUIZ

QUESTION 2 OF 3

Why use `while not agent.detect(BLOCK, FORWARD)` ?

So it keeps moving while there is no wall ahead.

So it never moves at all.

So it moves only when there IS a wall.

So it turns right forever.



QUIZ

QUESTION 3 OF 3

Why is a `while` loop better than 10 copies of `move forward` ?

It looks shorter only.

It stops exactly at the wall without guessing the number of steps.

It moves faster.

It places blocks too.

QUIZ

Your score

0 / 3



Re-read the slides. You'll get it.

↺ retake



Loop mastered!

Next: turn when you hit a wall.

다음 주: 벽을 만나면 방향 전환